AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT249
Practical modern statistics

Final Examination
Fall Semester 2024/2025

Date: 9 /Jan./2025

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nber of Exam Pages: 5 uding this cover sheet)	Time Allowed: 120 Minutes

structions:

- Total Marks: **100**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part1: Q-1 [20 marks]

A- [12 marks] Answer the following 5 questions as True or False (2 marks) and write the correct statement if the answer is false or state why it is false(1mark). Half (0.5) mark will be deducted for a wrong choice:

1- The diagonal elements of the correlation matrix are all equal to 0.
2- The time interval between cycles is the period.
3- In figure1, the principal component Y_2, shown on the y axis of the scatter plot, separates the groups very well.

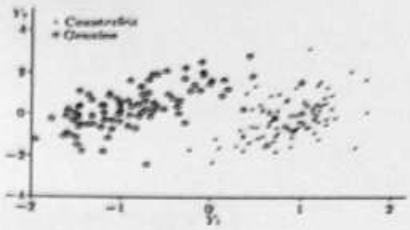
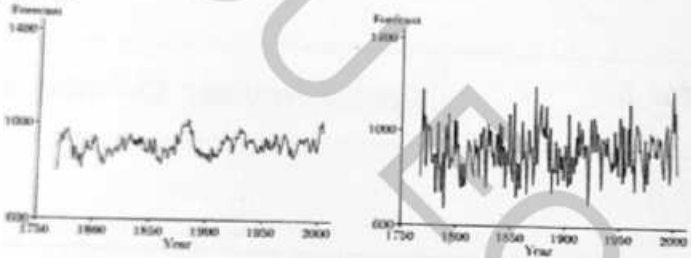


Figure 1

4- A matrix scatterplot is a series of two-dimensional scatterplots arranged in a grid.
5- Figure2 shows two sets of 1-step ahead forecasts using simple exponential smoothing, with two different values of the smoothing parameter α. The smoothing parameter α for figure2(a) is smaller than the smoothing parameter α for figure2 (b).

Figure 2



(a) (b)

B- [8 marks] Fill in the blanks in the following 8 statements:

1- The ..... of the moving average should be chosen so as to avoid both over-smoothing and under-smoothing the time series.
2- The ..... at lag k is a correlation coefficient r_k calculated between a time series and a copy of itself, lagged by k places.
3- Two approximations Y_1 and Y_2 to a multivariate data set are ..... if constants $c_1 \neq 0$ and c_2 can be found such that $Y_2 = c_1Y_1 + c_2$.
4- A ..... is a regular pattern that repeats at fixed intervals.
5- In a scree plot, the ..... is the point at which the plot flattens out.
6- A set of points lying close together on a ..... is often referred to as a cloud of points.
7- The number of variables, p, in a multivariate data set is the ..... of the data set.
8- On a scatterplot, two groups are said to be ....., if the areas of the scatterplot occupied by the two clouds of points corresponding to these groups do not overlap.

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Part2: Each question is worth 20 marks. Answer the following questions:

Q–2: [6+4+6+4 marks]:

In a study investigating reactions to immunizations, 42 young adults were given a combined injection of diphtheria and tetanus toxoids. Four weeks later, the level of antibodies that each adult had to each of these toxoids was recorded. A histogram of the combined scores for the 42 adults is shown in Figure 3.

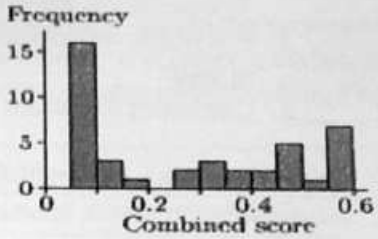


Figure **3**

(a) Identify the two variables recorded in this study.

(b) Use the histogram in Figure 3 to describe the distribution of combined antibody scores.

(c) Three of the participants had both local (such as redness near the injection site) and systemic (such as fever) adverse reactions (symptoms) to the injection. The recorded antibody levels to both diphtheria and tetanus for these three participants are given in Table1. The **mean diphtheria antibody** level is **1.63**, calculate the mean tetanus antibody level for these three participants.

Table 1 Antibody levels in three participants

Tetanus (X_1)	Diphtheria (X_2)
2.8	1.3
3.4	1.1
3.4	2.5

(d) Of the remaining 39 participants in the study, **nineteen** had no adverse reaction & their mean vector (1.52, 0.67), **thirteen** had a local adverse reaction only & their mean vector is (1.92, 0.94), and **seven** had a systemic reaction only & their mean vector is (2.84, 0.68).

Assuming that systemic reactions are more serious than local reactions, and that having both a systemic and a local reaction is worse than having a systemic reaction only, does there appear to be a relationship between mean antibody levels and strength of adverse reaction? If so, is the relationship linked with antibodies to both toxoids, or to just one? Justify your answer.

Q–3: [10+2+8 marks]:

(a) The matrix scatterplot for some multivariate data is shown in Figure 4. In this data set, the observations correspond to different calendar years, and the variables relate to various aspects of the weather at a particular location during the year.

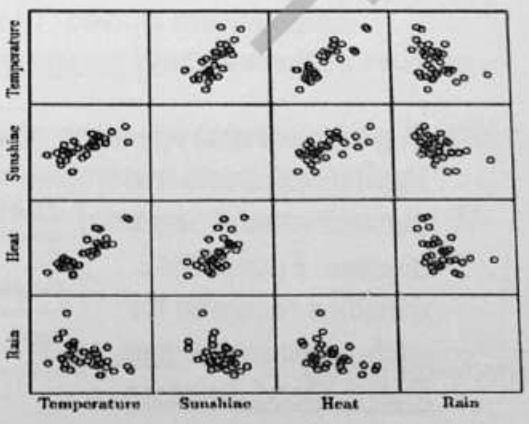


Figure 4

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1. Use Figure 4 to comment on the relationships between the variables Temperature, Sunshine, Heat and Rain.
2. For which two variables is the relationship strongest? Explain your answer.
3. Comment why there are no scatterplots in the diagonal positions.

(b) For the following matrix, state why it cannot be the correlation matrix for a set of data.

$$\begin{pmatrix} 1 & 0.2 & -0.3 & 0.7 \\ 0.2 & 1 & 0.4 & 0.5 \\ -0.3 & 0.4 & 1 & 0.9 \end{pmatrix}$$

(c) Four variables, X_1, X_2, X_3 & X_4, were recorded for a set of 12 observations. The correlation matrix, calculated using these data, is

$$\begin{pmatrix} 1 & 0.540 & 0.548 & 0.099 \\ 0.540 & 1 & 0.668 & 0.002 \\ 0.548 & 0.668 & 1 & 0.169 \\ 0.099 & 0.002 & 0.169 & 1 \end{pmatrix}$$

Use this correlation matrix to answer the following questions:

1. Between which pair of variables does there appear to be the strongest linear relationship? Between which pair of variables does there appear to be the weakest linear relationship? (In answering this question, you may assume that a correlation less than 0.20 indicates that there is little or no association.)
2. How is X_1 related to X_3?

Q–4: [5+5+10marks]:

(a) Descriptions of two multivariate data sets are given below. For each data set, state the dimension p and the number of observations n.

1. In a study on interest rates in different countries, data on fifteen countries was collected. For each country, three interest rates were recorded: the average six-month interest rate during 2004, the average three-month interest rate during 2004, and the average interest rate for ten-year government bonds in 2004.

2. A study was conducted to investigate the effect of exercise on the bones of people with osteoporosis. Samples of bone tissue from thirteen patients were studied. The levels of each of five proteins in each sample were measured.

(b) In a study of heart disease, measurements were taken on 120 patients with heart disease and 150 patients without heart disease. Four of the variables recorded for each patient were age, resting blood pressure,

Table 2

Variable	Heart disease present Mean	SD	Heart disease absent Mean	SD
Age	56.6	8.12	52.7	9.51

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serum cholesterol and maximum heart rate. The mean and standard deviation for the age variable for each group of patients are given in Table2

1. For the variables age, the grand mean is 54.4333, obtain the between-groups variance and the within-groups variance, and hence calculate the separation between patients with heart disease and patients without heart disease.
2. The separations for the other three variables are given in Table3 below

   Which variable gives the best separation between the two groups of patients? Which gives the least separation? Explain your answer.

Table 3

Variable	Separation
Blood pressure	0.024
Cholesterol	0.014
Maximum heart rate	0.211

Q–5: [5+15 marks]:

(a) A principal component analysis was carried out on a standardized data set consisting of 20 variables. The resulting scree plot is shown in Figure 5.
Based on the scree plot in Figure 5, how many components should be retained explain your answer.

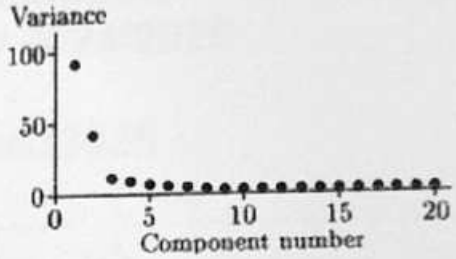


Figure 5

(b) There are five machines in a factory. Of these machines, (four) are working properly and (one) is defective. Machines which are working properly produce articles each of which has independently a probability of (0.1) of being imperfect. For the defective machines this probability is (0.3).
(six) articles are examined

1- Find the probability that (2) articles out of the (6) articles are imperfect given that the machine is defective.
2- Find the probability that (2) articles out of the (6) articles are imperfect given that the machine is not defective.

A machine is chosen at random and (six) articles produced by the machine are examined.

3- What is the probability that the machine chosen is defective given that, of the (six) articles examined, two are imperfect and (four) are perfect?
Let D be "2 out of 6 imperfects", let M be "machine defective" and let $\bar{M}$ be "machine not defective."

End of questions

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